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THERMALLY ADAPTIVE TOP FOIL OF A GAS FOIL BEARING

Abstract

A component of a gas foil bearing, having: a plurality of material layers forming a composition gradient that defines a first coefficient of thermal expansion (CTE) and a second CTE that differs from the first CTE, wherein: the plurality of material layers include at least two dissimilar metals, plastics, or fiber filled metals or plastics, that are layered on top of each other; when the component is subject to heating, the component changes from a first shape to a second shape; and the component is a top foil.

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Background/Summary

BACKGROUND

[0001] The embodiments are directed to gas foil bearings and more specifically to a thermally adaptive top foil of a gas foil bearing.

[0002] Thermal stability may impact the effectiveness of a of gas foil bearing (GFB) that supports an air cooled and air film supported rotating shaft. A GFB may operate continuously at high ambient temperatures. During normal operation, i.e., under nominal loads, an air film between the GFB and the shaft allows for both hydrodynamic pressure generation and heat flow in the bearing. The heat flow provides the desired level of temperature homogenization within the bearing body. A maximum value of a temperature gradient in the GFB is typically found in the region of a mechanical load transfer, due to the raised shear forces in the air film supporting the rotating shaft. If the bearing is overloaded, or during low rotational speed, the bearing clearance with the shaft may become thin which may result in a raised temperature gradient. The excessive gradients may lead to an uneven thermal expansion of the bearing structural components. Such thermally induced deformation and warping of the GFB geometry may also result in thermal stability loss, i.e., a blocking of internal bearing channels for airflow, or the shaft becoming stuck in the GFB outer shaft or bushing.

BRIEF DESCRIPTION

[0003] Disclosed is a component of a gas foil bearing, including: a plurality of material layers forming a composition gradient that defines a first coefficient of thermal expansion (CTE) and a second CTE that differs from the first CTE, wherein: the plurality of material layers include at least two dissimilar metals, plastics, or fiber filled metals or plastics, that are layered on top of each other; when the component is subject to heating, the component changes from a first shape to a second shape; and the component is a top foil.

[0004] In addition to one or more aspects of the component or as an alternate, the first shape is a curved triangle.

[0005] In addition to one or more aspects of the component or as an alternate, the second shape is circular.

[0006] In addition to one or more aspects of the component or as an alternate, the plurality of layers include: a first layer that forms a body of the component; and a second layer that has a same size and shape as the first layer and extends about an exterior side or an interior side of the first layer.

[0007] In addition to one or more aspects of the component or as an alternate, the plurality of layers include: a first layer forms a body of the component; and a second layer extends about a portion of an exterior side or an interior side of the first layer.

[0008] In addition to one or more aspects of the component or as an alternate: when the component forms the curved triangle, portions of the first layer define corners of the curved triangle; and the second layer forms discrete segments that are located at the corners of the curved triangle.

[0009] In addition to one or more aspects of the component or as an alternate, the second layer has a thickness of 10 microns or less.

[0010] In addition to one or more aspects of the component or as an alternate, the second layer has a thickness of 10 microns or less.

[0011] Further disclosed is a gas foil bearing, including: a component having one or more aspects disclosed herein; a bump foil surrounding the component; and bearing sleeve surrounding the bump foil. In addition to one or more aspects of the gas foil bearing or as an alternate, the bearing further includes: a journal shaft surrounded by the component and supported by the bearing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0013] FIG. 1 shows an aircraft according to an embodiment;

[0014] FIG. 2 shows a gas foil bearing (GFB) that may be utilized in the aircraft, according to an

embodiment;

[0015] FIG. **3** shows a top foil of the GFB in one shape state, when the GFB is relatively warm;

[0016] FIG. **4** shows the top foil in a second shape state, when the GFB is relatively cold;

[0017] FIG. **5** shows a detail of the top foil, utilizing material layers having different coefficients of thermal expansion (CTEs) that result in the predetermined shape states;

[0018] FIG. **6** shows an element that is equivalent to the top foil configurations disclosed herein, with two material layers having different CTEs, at a temperature T_1 ;

[0019] FIG. **7** shows the structure of FIG. **56** at a temperature $T_2 > T_1$; and

[0020] FIG. **8** shows an element similar to that of FIG. **6** at a temperature $T_2 > T_1$, where a material interface allows free slip between layers, such the layers extend at different rates upon being heated.

DETAILED DESCRIPTION

[0021] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0022] FIG. **1** shows an aircraft **1** having a fuselage **2** with a wing **3** and tail assembly **4**, which may have control surfaces **5**. The wing **3** may include an engine **6**, such as a gas turbine engine, and an auxiliary power unit **7** may be disposed at the tail assembly **4**. The aircraft **1** may have an air cycle machine **10**.

[0023] Turning to FIG. **2**, rotating parts such as a journal shaft (or shaft) **100** may rotate within a gas foil bearing (GFB) **110**. The GFB **110** includes a bearing sleeve **120** or shell, or bushing, that may be cylindrical. The bearing sleeve **120** may surround a bump foil **130**, which may be ring shaped with an undulating profile. The bump foil **130** may surround a top foil **140**, which also may be ring shaped and have a cylindrical profile. A predetermined gap **150** may be defined between the top foil **140** and the shaft **100**, which may be separated by a layer of air film during rotation of the shaft **100**.

[0024] FIG. **3** schematically shows the shape of the top foil **140**, as a closed loop for simplicity, at a first temperature T_1 . FIG. **4** shows the shape of the top foil **140** at a second temperature $T_2 < T_1$. That is, in a relatively cold state of the GFB **110**, the top foil **140** has a curved triangular shape so that there are at least three points of contact **155A-155C** between the GFB **110** and the internal shaft **100** (e.g., a first shape state). When the temperature of the GFB **110** is elevated, e.g., during steady-state operation, the shape of the top foil **140** is round (e.g., a second shape state).

[0025] To obtain the shape changes shown in FIGS. **3** and **4**, as shown in FIG. **5**, the top foil **140** may be additively manufactured with a plurality of dissimilar materials, including dissimilar metals or thermoplastic polymers which may be filled with dissimilar fibers. Where two or more metals are utilized in the disclosed embodiments, the metals may include ones that form memory alloys in the regions of mixed composition. Alternatively, a memory alloy may be utilized to manufacture the foil **140**. With the materials, the top foil **140** forms a composition gradient that defines a first coefficient of thermal expansion (CTE) and a second CTE that differs from the first CTE. With the composition gradient, the different shapes shown in FIGS. **3** and **4** may be obtained at different operating temperatures of the GFB **110**.

[0026] More specifically the top foil **140** has a first layer **158** of a material forming a body **160** of the top foil **140**. The first layer **158** may have a first coefficient of thermal expansion (CTE). By utilizing additive manufacturing, a second layer **170** of a second material may be provided around a portion the body **160** or the entire body **160** of the top foil **140**, i.e., having a same size and shape as the first layer **158**. The second layer **170** may be **10** microns thick. The second material may be a different material than the first material, having a second CTE that differs from the first CTE to form a CTE gradient.

[0027] In one embodiment, the second layer is distributed only at predetermined locations **180**, such as three portions **180A-180C** of the first layer **158**, which are circumferentially distributed around the body **160** of the top foil **140**. These locations **180A-180C** form corners of the curved

triangular shape shown in FIG. 4. In one embodiment the second layer **170** is formed on an exterior side **160A** of the bearing sleeve body **160**, though the second layer **170** may be formed on an interior side **160B** of the bearing sleeve body **160**.

[0028] FIGS. **6** and **7** show a structural element **248** that is equivalent to the layers shown in FIG. **6**. Specifically, first and second layers **200**, **210**, which may be dissimilar metals, thermoplastic polymers, and which may be filled with dissimilar fibers, to provide dissimilar CTEs. The layers **200**, **210** are at a temperature T_1 in FIGS. **6**, and T_2 that is greater than T_1 in FIG. **7**. The controlled thermal expansion shown in FIG. **7** results from the layers **200**, **210** being integrally connected. That is, the first and second layer **200**, **210** bend together in a predictable and controlled way. That is, the controlled thermal expansion of the first and second layers **200**, **210** provides the desired shape change of the structural element **248**. FIG. **8** shows an element **405** that is similar to that of FIG. **7** at a temperature $T_2 > T_1$, where a material interface allows free slip between layers **450**, **460**, such the layers **450**, **460** extend at different rates upon being heated, rather than deforming as shown in FIG. **6**.

[0029] Benefits of a top foil **140** that deforms as shown in FIGS. **3** and **4** include a reduction of non-uniform or undesired deformation that may otherwise occur with temperature gradients that are induced throughout the operation of the GFB **110**. That is, the embodiments provide a GFB **110** that has an optimized stiffness over substantially an entire range of operation of the GFB **110**. The GPF **110** that deforms as shown in FIGS. **3-4** may increase the bearing capacity, reduce startup drag, increase durability, and provide improved modal stability. In addition, the embodiments may reduce startup drag and wear.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0031] Those of skill in the art will appreciate that various example embodiments are shown and described herein, each having certain features in the particular embodiments, but the present disclosure is not thus limited. Rather, the present disclosure can be modified to incorporate any number of variations, alterations, substitutions, combinations, sub-combinations, or equivalent arrangements not heretofore described, but which are commensurate with the scope of the present disclosure. Additionally, while various embodiments of the present disclosure have been described, it is to be understood that aspects of the present disclosure may include only some of the described embodiments. Accordingly, the present disclosure is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

Claims

1. A component of a gas foil bearing, comprising: a plurality of material layers forming a composition gradient that defines a first coefficient of thermal expansion (CTE) and a second CTE that differs from the first CTE, wherein: the plurality of material layers include at least two dissimilar metals, plastics, or fiber filled metals or plastics, that are layered on top of each other; when the component is subject to heating, the component changes from a first shape to a second shape; and the component is a top foil.
2. The component of claim 1, wherein the first shape is a curved triangle.
3. The component of claim 2, wherein the second shape is circular.
4. The component of claim 3, wherein the plurality of layers include: a first layer that forms a body of the component; and a second layer that has a same size and shape as the first layer and extends

about an exterior side or an interior side of the first layer.

5. The component of claim 3, wherein the plurality of layers include: a first layer forms a body of the component; and a second layer extends about a portion of an exterior side or an interior side of the first layer.

6. The component of claim 5, wherein: when the component forms the curved triangle, portions of the first layer define corners of the curved triangle; and the second layer forms discrete segments that are located at the corners of the curved triangle.

7. The component of claim 4, wherein the second layer has a thickness of **10** microns or less.

8. The component of claim 5, wherein the second layer has a thickness of 10 microns or less.

9. A gas foil bearing, comprising: the component of claim 1; a bump foil surrounding the component; and bearing sleeve surrounding the bump foil.

10. The bearing of claim 9, further comprising: a journal shaft surrounded by the component and supported by the bearing.
